

Zhaodong Liu

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EDUCATION

Carnegie Mellon University – Pittsburgh, PA Dec 2027(Expected)
Master of Computational Data Science (MCDS): Analytics Track
New York University Shanghai – Shanghai, China May 2026
Bachelor of Science; Double Major: Computer Science, Mathematics, Cum Laude
Cumulative GPA: **3.793/4.0** | CS Major GPA: **3.946/4.0** | Math Major GPA: **3.786/4.0**

SKILLS

Programming: Python, Java, C, SQL, R | **ML & AI:** PyTorch, LoRA/PEFT, Hugging Face, Transformers
Data & Tools: NumPy, Pandas, MySQL, Git | **Languages:** Chinese (native), English (fluent)

PROFESSIONAL EXPERIENCE

Data Engineer Intern, Data Strategy Department, McDonald's China Dec 2025 - Apr 2026

- Engineered an internal AI agent for self-service data access, routing natural-language queries to constraint-aware SQL over a catalog spanning 14 business domains, or RAG over internal data maps and documents to generate cited answers.
- Developed a similarity-based sales forecasting function over a combo master-data set normalized across 4 ID levels, quantifying menu-composition to forecast new-combo launch sales from comparable combos.
- Managed and validated enterprise database schemas, governing a 203-scenario and 54-interface data map.
- Redesigned data dashboards, organizing 500+ financial metrics with structured drill-down paths for self-service analytics.

RESEARCH EXPERIENCE

Research Assistant, New York University Shanghai May 2025 - Jan 2026
Generative Retrieval for Multimodal Recommendation System - Advisor: Prof. Hongyi Wen

- Developed a multimodal framework for sequential recommendation on Amazon datasets. Identified and applied VQ-VAE, RQ-VAE, OPQ to capture discrete behavior patterns, achieving 47.2% improvement in NDCG@10 (0.0339 to 0.0499).
- Implemented BPE-inspired partitioning tokenization across item sequences, enhancing coherence and representation quality.

Research Assistant, New York University Shanghai May 2024 - Aug 2024
Energy-Aware Route Optimization Under Traffic-Light Timing - Advisor: Prof. Zhibin Chen

- Proposed and designed a traffic-aware routing system under real-world map constraints, minimizing energy consumption.
- Built an OSM, SUMO, TraCI pipeline and independently extended Dijkstra's algorithm to multi-objective (time, energy) shortest paths, integrating electric and gasoline vehicle energy models. Achieved 10–15% fuel savings at 2–5% time cost.

PROJECTS

Finetuning on Pretrained LM for Efficient Retrosynthesis of Chemical Reactions Sep 2024 - Dec 2024
Center for Data Science, New York University

- Trained predictive models for chemical reaction retrosynthesis using SMILES notation, applying an LLM (LlaSMol-Llama2) for dataset reconstruction and augmentation, improving Top-1 Accuracy by 26.2% (0.588 to 0.742).
- Employed fine-tuning strategies (encoder freezing, MLP adapters, LoRA) to optimize training efficiency.

Fantasy Sports League Database Sep 2024 - Dec 2024
Tandon School of Engineering, New York University

- Self-directed a project to design a 13-table relational database in MySQL supporting multi-league sports with user authentication, draft scheduling, roster management, and trading systems. Engineered a Python backend and HTML/CSS frontend with ACID-compliant transactions, triggers, and stored procedures, integrating full-stack functionality.

HONORS

New York University Shanghai, Founders' Day Award May 2026
New York University Shanghai, Dean's List Sep 2022 - May 2025
Mathematical Contest in Modeling, Certificate of Achievement, Honorable Mention May 2023

- Preprocessed large datasets and built predictive models, including Bayesian regression, for regional market data analysis.

LEADERSHIP

Trombone Section Leader, Chamber Orchestra, New York University Shanghai Sep 2022 - May 2026

- Directed trombone section, coordinated weekly rehearsals and semester concerts, and oversaw instrument storage logistics.

Treasurer, PEER-Pal Club, New York University Shanghai May 2023 - Aug 2024

- Managed financial operations and budgeting; facilitated letter exchanges with high school students from Hunan Province.